CMO Qualifying Repechage

My Solutions to Problems 1–7

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Problems

1. **[10 points]** Do there exist four lines on the same plane, no three concurrent and no two parallel, so that four particles can travel at (possibly different) constant non-zero speeds along these lines and such that at every point in time, the positions of the four particles are concyclic?

2. **[10 points]** Do there exist positive real numbers a, b, c, not all equal, such that $a^b = b^c = c^a$?

3. **[10 points]** Let point P be outside circle Γ. The tangents from P to Γ hit Γ at A and B. A third line through P hits Γ at C and D, such that C is between P and D. Point Q is on chord CD such that $\angle DAQ = \angle PBC$. Prove that $\angle DBQ = \angle PAC$.

4. **[15 points]** Fix a prime p and natural numbers $k \geq n$. Consider choosing k vectors $v_1, \ldots, v_k$ uniformly at random (possibly with repetition) from the set $\{(x_1, \ldots, x_n) \mid x_i \in \{0, 1, \ldots, p-1\}\}$. Also, let v_{k+i} be the vector with value p in the i-th coordinate and 0 elsewhere for $1 \leq i \leq n$.

   These $k+n$ vectors are said to *generate* $\mathbb{Z}^n$ if for any vector of integers $v = (y_1, \ldots, y_n)$, there are some integers $c_1, \ldots, c_{k+n}$ such that $v = \sum_{i=1}^{k+n} c_i v_i$. Let $P(p, n, k)$ be the probability that these $k + n$ vectors generate $\mathbb{Z}^n$.

   Then, there is some constant c, depending on p and n, such that $p^k(1 - P(p, n, k)) - c$ approaches 0 as $k \to \infty$. Determine c in terms of p and n.

5. **[15 points]** Determine whether there exists a surjective function $f : (0, \infty) \to \mathbb{N}$, such that for any positive numbers $a < b$, the image of the open interval (a, b) under f is a set of the form $\{1, 2, \ldots, N\}$ for some finite positive integer N.

   Recall that a function $f : (0, \infty) \to \mathbb{N}$ is *surjective*, if for every natural number M, there exists some positive real number r such that $f(r) = M$.

6. **[20 points]** A set of positive integers T is given. For a positive integer n, Alice and Bob play a game, where there are initially n stones in a pile, they take turns taking m stones from the pile, where $m \in T, m \leq n$, and whoever takes the last stone wins. If at some point the number of stones in the pile is less than $\min T$, then the game is a draw. Call a number n *good* if Alice can guarantee a win, *bad* if Bob can guarantee a win, and *neutral* if neither Alice nor Bob can guarantee a win.

   For example, if T is the set of prime numbers, then $n = 4$ is neutral and $n = 5$ is good; in the first game, Alice can guarantee a draw by taking 3 stones on her first move, and cannot do better, and in the second game, she can take all the stones on her first move to win.

   (a) Prove that if T is finite, then the set of good numbers is eventually periodic: that is, there exists some positive integers N and p such

that for all positive integers $n > N$, we have that n is good if and only if $n + p$ is good.

(b) Construct an infinite set T such that the set of good numbers is not eventually periodic: that is, for every pair of positive integers N and p, there exists some $n > N$ such that exactly one of n and $n + p$ is good.

[*note*: the word 'eventually' was originally missing from this question part b but we added it and notified students during the exam.]

7. **[20 points]** Determine the least positive integer that is not a perfect square that can be written in the form $\dfrac{a^2 + b^2}{ab + 1}$ for some rational numbers a and b such that $ab \neq -1$.

Problem 1

Solution: Let ω be a circle with centre O and radius r. Let A, B, C, D be four distinct points on w. Let l_A, l_B, l_C, l_D be the lines tangent to ω through A, B, C, D, respectively, and suppose that no two of these lines are parallel. Since there are at most 2 lines tangent to ω passing through any point P in the plane, we know that no three of l_A, l_B, l_C, l_D are concurrent. We claim that if four particles begin at A, B, C, D and move along l_A, l_B, l_C, l_D respectively, clockwise in positive time at the same speed $v \neq 0$, they are always concyclic. Let the position of the particles after time t (not necessarily positive)

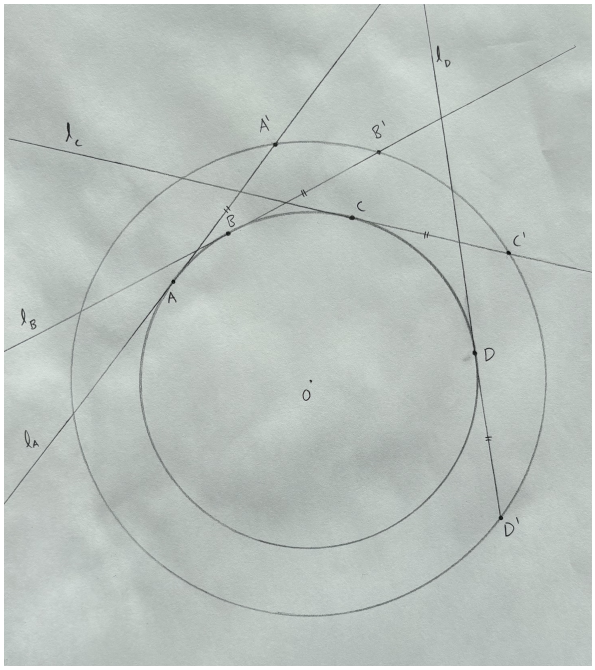


Figure 1: Diagram for Problem 1

be A', B', C', D'. Since the lines l_X are all tangent to ω, we have $\angle OXX' = 90°$ for $X = A, B, C, D$. In time t the particles all travel a distance of vt, thus

$$\triangle OAA' \cong \triangle OBB' \cong \triangle OCC' \cong \triangle ODD'$$

by SAS congruence. Hence $OA' = OB' = OC' = OD'$, so the particles are always equidistant from O. □

Problem 2

Solution: We claim that there are no such numbers. Assume on the contrary that positive real numbers a, b, c are not all equal and satisfy the given condition.

Claim: None of a, b, c is equal to 1.

Proof. Without loss of generality, if $a = 1$, then

$$1 = b^c = c \implies 1 = a = b = c,$$

which is not permitted. □

Claim: We must have a, b, c be pairwise distinct.

Proof. Suppose without loss of generality that $a = b$. Then by the given condition

$$\begin{aligned} a^b = b^c &\implies a^a = a^c \\ &\implies a = c, \end{aligned}$$

so all three are equal, a contradiction. □

Since the given condition is cyclic but not symmetric in a, b, c, we consider the two cases for their ordering up to a cyclic permutation of the variables: either $a > b > c$ or $c > b > a$. Note that these inequalities are strict due to the previous claim. We consider each case separately, and show that they both lead to contradictions. In each case, we use the following lemma.

Lemma 1. *Let x, y, z be positive real numbers, and suppose that $x > y$. Then we have*

1. $x^z > y^z$,
2. $z > 1$ *implies* $z^x > z^y$,
3. $z < 1$ *implies* $z^y > z^x$.

Proof. For the first statement, note that $a^z > 1$ for all $a > 1$. Thus

$$\begin{aligned} x > y &\implies \frac{x}{y} > 1 \\ &\implies \frac{x^z}{y^z} > 1 \\ &\implies x^z > y^z. \end{aligned}$$

For part 2, note that if $z > 1$, $x - y > 0$ implies that $z^{x-y} > 1$. Hence

$$z^y(z^{x-y} - 1) > 0 \implies z^x > z^y.$$

For part 3, we instead have $z^{x-y} < 1$, since $z < 1$. As a result,

$$z^y(z^{x-y} - 1) < 0 \implies z^y > z^x.$$

□

Now we proceed with the casework.

Case 1 ($a > b > c$)**.** By the first claim, $a \neq 1$. If $a > 1$, then by the lemma, $a^b > a^c$ and $a^c > b^c$, hence $a^b > b^c$. This is a contradiction since it was assumed these two quantities are equal.

Now if $1 > a$, we must have $1 > b$ as well. By the lemma, $b^c > b^a$ and $b^a > c^a$. Combining these gives $b^c > c^a$, once again a contradiction.

Case 2 ($c > b > a$)**.** This time, we consider the position of both c and b relative to 1. First, suppose $c, b > 1$. Then we have $b^c > b^b$ and $b^b > a^b$. Combining these gives the contradiction $b^c > a^b$.

Now suppose $c > 1$ and $b < 1$. We have $c^a > b^a$ and $b^b > a^b$, but since $b < 1$, we also have $b^a > b^b$. Therefore, $c^a > b^a > b^b > a^b$, a contradiction.

Finally, suppose that $c < 1$. Our lemma gives us $c^a > c^b$ and $c^b > a^b$. Hence $c^a > a^b$, which is a contradiction. □

Problem 3

Solution: Notice that $\overline{CD}$ is the D-symmedian of $\triangle ABD$ and that quadrilateral $ACBD$ is harmonic. Since line BP is tangent to Γ at B, we have

$$\angle DAQ = \angle PBC = \angle CAB.$$

Evidently, $\overline{AB}$ is isogonal to $\overline{AQ}$ with respect to $\triangle ACD$. Since $ACBD$ is harmonic, $\overline{AB}$ is the A-symmedian of $\triangle ACD$; it follows that Q is the midpoint of $\overline{CD}$. Similarly, $\overline{AB}$ is the B-symmedian of $\triangle BCD$. Hence $\overline{AB}$ is isogonal to $\overline{BQ}$ with respect to $\triangle BCD$, so

$$\angle DBQ = \angle ABC = \angle PAC,$$

as desired. □

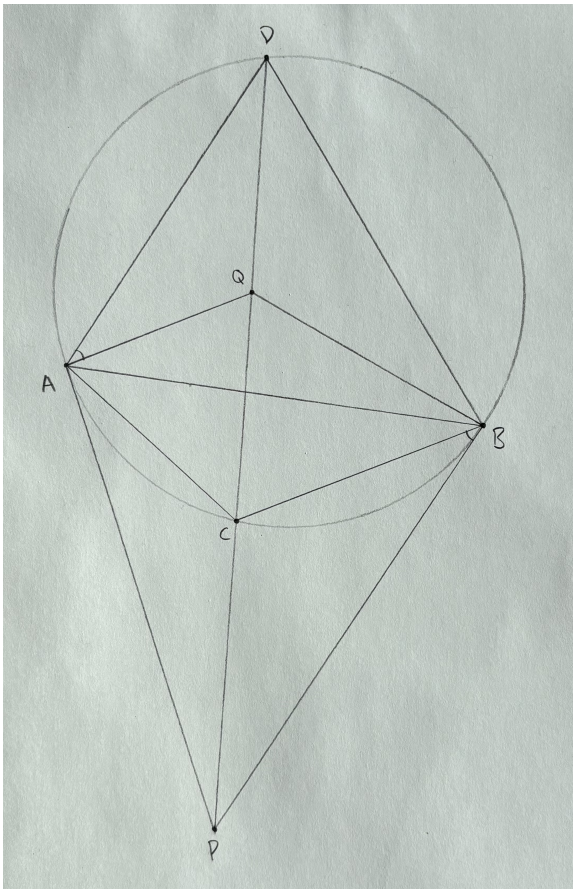


Figure 2: Diagram for Problem 3

Problem 4

Solution: We claim the answer is $\frac{p^n-1}{p-1}$. To prove this, we first show the following.

Claim: The $k+n$ vectors generate $\mathbb{Z}^n$ if and only if the k random vectors generate $(\mathbb{Z}/p\mathbb{Z})^n$.

Proof. First, suppose the $k+n$ vectors generate $\mathbb{Z}^n$. Let $\mathbf{v}$ be a vector in $(\mathbb{Z}/p\mathbb{Z})^n$. Since the $k+n$ vectors generate $\mathbb{Z}^n$, there exists some integers $c_1, \ldots, c_{k+n}$ such that

$$\begin{aligned}\mathbf{v} &= \sum_{i=1}^{k+n} c_i \mathbf{v}_i \\ &= \sum_{i=1}^{k} c_i \mathbf{v}_i + \sum_{i=k+1}^{k+n} c_i \mathbf{v}_i.\end{aligned}$$

However, all of the vectors in the second sum are the zero vector in $(\mathbb{Z}/p\mathbb{Z})^n$. Thus in $(\mathbb{Z}/p\mathbb{Z})^n$, we have $\mathbf{v} = \sum_{i=1}^{k} c_i \mathbf{v}_i$. Hence the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ generate $(\mathbb{Z}/p\mathbb{Z})^n$.

Conversely, suppose that the k random vectors generate $(\mathbb{Z}/p\mathbb{Z})^n$. Let $\mathbf{w} = (y_1, \ldots, y_n)$ be a vector in $\mathbb{Z}^n$. We know that in $(\mathbb{Z}/p\mathbb{Z})^n$, $\mathbf{w}$ can be written as a linear combination of the k random vectors. Thus this same linear combination in $\mathbb{Z}^n$ will produce a vector $(x_1, \ldots, x_n)$ with each coordinate x_i differing from y_i by a multiple of p. Let $y_i - x_i = pq_i$ for each $1 \leq i \leq n$. Then the linear combination $q_1\mathbf{v}_{k+1} + \cdots + q_n\mathbf{v}_{k+n}$ will correct for this difference. Hence every vector in $\mathbb{Z}^n$ can be written as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_{k+n}$, so these vectors generate $\mathbb{Z}^n$. □

Having shown this claim, we may disregard $\mathbf{v}_{k+1}, \ldots, \mathbf{v}_{k+n}$, instead working with the k random vectors in $(\mathbb{Z}/p\mathbb{Z})^n$.

Note that the k random vectors *not* generating $(\mathbb{Z}/p\mathbb{Z})^n$ is equivalent to the dimension of the spanning set of these vectors being less than or equal to $n-1$. This is equivalent to the vectors being contained within some $n-1$ dimensional subspace of $(\mathbb{Z}/p\mathbb{Z})^n$. Hence we have

$$\begin{aligned}1 - P(p,n,k) &= 1 - P(\mathbf{v}_1, \ldots, \mathbf{v}_k \text{ generate } (\mathbb{Z}/p\mathbb{Z})^n) \\ &= P(\mathbf{v}_1, \ldots, \mathbf{v}_k \text{ in some } n-1 \text{ dimensional subspace of } (\mathbb{Z}/p\mathbb{Z})^n)\end{aligned}$$

Claim: The number of $n-1$ dimensional linear subspaces of $(\mathbb{Z}/p\mathbb{Z})^n$ is $\frac{p^n-1}{p-1}$.

Proof. If $n = 1$, this is trivially true; there is just one zero-dimensional subspace, namely the one containing only the zero vector. This matches $\frac{p^1-1}{p-1} = 1$.

Thus suppose $n \geq 2$. We construct a $n-1$ dimensional linear subspace by picking $n-1$ linearly independent vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{n-1}$ as its basis. The first

such vector $\mathbf{v}_1$ only has to be nonzero, so there $p^n - 1$ choices for $\mathbf{v}_1$. Each subsequent vector $\mathbf{v}_i$ that we pick cannot be one of the p^{i-1} vectors in the span of the previous ones, so there are $p^n - p^{i-1}$ choices for $\mathbf{v}_i$. This gives us an initial count of

$$(p^n - 1)(p^n - p) \dots (p^n - p^{n-2})$$

subspaces. However, since each $n-1$ dimensional subspace can be generated by multiple different bases, this overcounts. We must divide our count by the number of bases $(\mathbf{w}_1, \dots, \mathbf{w}_{n-1})$ that generate any given $n-1$ dimensional subspace. Counting this is done much in the same manner as above; there are $p^{n-1} - p^{i-1}$ choices for each $\mathbf{w}_i$, meaning we overcounted by a factor of

$$(p^{n-1} - 1)(p^{n-1} - p) \dots (p^{n-1} - p^{n-2}).$$

Taking the quotient between our original count and overcount factor gives us the number of $n-1$ dimensional linear subspaces:

$$\begin{aligned}\frac{(p^n - 1)(p^n - p) \dots (p^n - p^{n-2})}{(p^{n-1} - 1)(p^{n-1} - p) \dots (p^{n-1} - p^{n-2})} &= \frac{(p^n - 1)(p^{n-1} - 1) \dots (p^2 - 1)}{(p^{n-1} - 1)(p^{n-2} - 1) \dots (p - 1)} \\ &= \frac{p^n - 1}{p - 1},\end{aligned}$$

as desired. □

Let the m linear subspaces of dimension $n-1$ in $(\mathbb{Z}/p\mathbb{Z})^n$ be $A_1, A_2, \dots, A_m$, where $m = \frac{p^n-1}{p-1}$. Let $A_{i_1,i_2,\dots,i_r}$ denote the intersection of subspaces $A_{i_1}, \dots, A_{i_r}$. Further, let $P(A_{i_1,i_2,\dots,i_r})$ be the probability that the k random vectors are contained in $A_{i_1,i_2,\dots,i_r}$. We now use the principle of inclusion-exclusion to determine the probability that the randomly selected vectors are contained in *some* A_i. We have

$$\begin{aligned}p^k(1 - P(p,n,k)) = {} & p^k(P(A_1) + \dots + P(A_m)) \\ & - p^k(P(A_{1,2}) + P(A_{1,3}) + \dots + P(A_{m-1,m})) \\ & + \dots \\ & - (-1)^m p^k P(A_{1,2,\dots,m}).\end{aligned} \tag{1}$$

Note that $P(A_i) = (\frac{p^{n-1}}{p^n})^k = \frac{1}{p^k}$ for all $1 \le i \le m$. To deal with the other terms, we use the following claim.

Claim: The intersection of two or more distinct $n-1$ dimensional linear subspaces of $(\mathbb{Z}/p\mathbb{Z})^n$ must contain strictly fewer than p^{n-1} elements.

Proof. Since the subspaces have dimension $n-1$, they both contain p^{n-1} elements. Thus their intersection has at most p^{n-1} elements. If their intersection has exactly p^{n-1} elements, then all the subspaces must be identical. However they were given to be distinct, so their intersection has strictly fewer than p^{n-1} elements. □

Therefore, if $r \geq 2$, we have

$$\begin{aligned} p^k P(A_{i_1,i_2,\ldots,i_r}) &< p^k \left(\frac{p^{n-1}}{p^n}\right)^k \\ &= 1 \\ \implies \lim_{k\to\infty} p^k P(A_{i_1,i_2,\ldots,i_r}) &= 0. \end{aligned}$$

This means that when we take the limit as $k \to \infty$ in (1), all of the terms after the first line, of which there are finitely many, will go to zero! Hence

$$\begin{aligned} \lim_{k\to\infty} p^k(1 - P(p,n,k)) &= \lim_{k\to\infty} p^k(P(A_1) + \cdots + P(A_m)) \\ &= \lim_{k\to\infty} p^k \left(\frac{m}{p^k}\right) \\ &= \lim_{k\to\infty} m \\ &= \frac{p^n - 1}{p-1}. \end{aligned}$$

Thus $\lim_{k\to\infty} p^k(1 - P(p,n,k)) - \frac{p^n-1}{p-1} = 0$, as desired. □

Problem 5

Solution: We construct a function $f : (0, \infty) \to \mathbb{N}$ that satisfies the given criteria. First, we define $g(q)$ to be the number of primes less than or equal to q, for any prime q. Hence $g^{-1}(n)$ is the nth prime, for any natural number n.

Further, let $p(x)$ be the smallest prime that divides the denominator of a fraction x when x is written as a fraction in lowest terms. We define the domain of p to be the set of all positive rational numbers that are not integers.

We now define f as follows.

$$f(x) = \begin{cases} g(p(x)), & \text{if } g(p(x)) < x \text{ and } x \in \text{dom}(p), \\ 1, & \text{otherwise.} \end{cases}$$

The following are some examples of f:

- $f(5) = 1$, since 5 is an integer.
- $f(100\pi) = 1$, since 100π is irrational.
- $f(\frac{410}{91}) = 4$, since $p(\frac{410}{91}) = 7$, and $g(7) = 4 < \frac{410}{91}$.
- $f(\frac{310}{91}) = 1$, since $p(\frac{310}{91}) = 7$, and $g(7) = 4 > \frac{310}{91}$.

Claim: The function f is surjective.

Proof. Let $M \in \mathbb{N}$. We claim that $f(r) = M$, where

$$r = \frac{Mg^{-1}(M) + 1}{g^{-1}(M)}.$$

To see why, note that r in the domain of p and that the fraction above is in lowest terms. Thus $p(r) = g^{-1}(M)$, and so

$$g(p(r)) = g(g^{-1}(M)) = M < M + \frac{1}{g^{-1}(M)} = r.$$

By the definition of f, it follows that $f(r) = g(p(r)) = M$. □

Claim: For any positive numbers $a < b$, the image of the open interval (a, b) under f is a set of the form $\{1, 2, \ldots, N\}$ for some finite positive integer N.

Proof. If $b \leq 2$, then $(a, b) \subset (0, 2]$. Suppose that $f(y) \neq 1$ for some $y \in (0, 2]$. Then y must be a rational noninteger such that $g(p(y)) = f(y) \geq 2$ and $g(p(y)) < y$. Hence $y > 2$; however this is a contradiction since $y \leq 2$. Therefore, $f(x) = 1$ for all $x \in (0, 2]$. Since $(a, b) \subset (0, 2]$ and $(a, b) \neq \emptyset$, the claim holds with $N = 1$.

Now suppose that $b > 2$. We show that the claim holds with $N = \lceil b \rceil - 1$.

Let $x \in (a, b)$. If $f(x) = 1$, we have $f(x) = 1 < \lceil b \rceil - 1 = N$. Otherwise, we must have $f(x) = g(p(x)) < x$, and since $f(x)$ is an integer, $f(x) \leq \lfloor x \rfloor$. Now if b is an integer, then

$$\begin{aligned} x < b \implies \lfloor x \rfloor &\leq b - 1 \\ &= \lceil b \rceil - 1 \\ &= N. \end{aligned}$$

If b is not an integer, we have $\lfloor x \rfloor \leq \lfloor b \rfloor = \lceil b \rceil - 1 = N$. Thus in all cases, $f(x) \leq N$. From the definition of f, we know $f(x) \geq 1$, hence $f(x) \in \{1, 2, \ldots, N\}$.

Now let $n \in \{1, 2, \ldots, N\}$. We will show that there must exist $r \in (a, b)$ such that $f(r) = n$. Let $k = \max(a, N)$. Since

$$\begin{aligned} \lceil b \rceil = N + 1 &\implies b + 1 > N + 1 \\ &\implies b > N, \end{aligned}$$

we have $b > k \geq N$.

Let $P = g^{-1}(n)$. We select an integer q sufficiently large such that $P^q > \frac{2}{b-k}$. Since $b > k$, both quantites are positive. Hence

$$bP^q - kP^q = (b - k)P^q > (b - k)\frac{2}{b - k} = 2.$$

Therefore, there exists at least two consecutive integers in the interval (kP^q, bP^q). Since P is prime, at least one of these integers must be coprime with P. Let one such integer be m; then

$$\begin{aligned} kP^q < m < bP^q &\implies k < \frac{m}{P^q} < b \\ &\implies \frac{m}{P^q} \in (a, b). \end{aligned}$$

The number $\frac{m}{P^q}$ is neither an integer nor irrational, so we consider $g(p(\frac{m}{P^q}))$. Clearly $p(\frac{m}{P^q}) = P$, so we have

$$\begin{aligned} g\left(p\left(\frac{m}{P^q}\right)\right) &= g(P) \\ &= g(g^{-1}(n)) \\ &= n, \end{aligned}$$

and further

$$n \leq N \leq k < \frac{m}{P^q}.$$

Therefore, $f(\frac{m}{P^q}) = n$, so the claim is proved. □

Problem 6

Solution:

a. Let $T = \{a_1, \ldots, a_k\}$ where all the a_i are positive integers, and $a_i < a_j$ for all $i < j$. Define $f(n)$ as the status (good/bad/neutral) of a positive integer n.

Claim: Let $n > a_k$. For all integers $m \geq n$, $f(m)$ is a function of the values of $f(i)$ for $n-a_k \leq i \leq n-a_1$. We call $f(n-a_k), f(n-a_k+1), \ldots, f(n-a_1)$ the *determining sequence* of n.

Proof. We prove this by induction, using n as our base case. Note that

- n is good if and only if there exists some i such that $n - a_i$ is bad, or $n \in T$,
- n is bad if and only if $n - a_i$ is good for all $a_i < n$, and $n \notin T$,
- n is neutral otherwise.

This gives us an algorithm based on the determining sequence to find $f(n)$, and it does not require knowing any other values of f. Thus $f(n)$ is uniquely determined by its determining sequence.

Now let $m > n$, and suppose the claim holds for $n, n+1, \ldots, m-1$. By the same logic as was just used, $f(m)$ is a function of its determining sequence. However, since $m > n$, any element of the determining sequence of m is either a subset of the determining sequence of n or, by the inductive hypothesis, a function of the determining sequence of n. Hence the claim holds for m, completing the induction. □

Claim: There exist two distinct positive integers with the same determining sequence.

Proof. Since T is finite, every determining sequence has a whole number of elements, specifically, $a_k - a_1 + 1$ of them. Since f only has three outputs in its range, there are at most $3^{a_k - a_1 + 1}$ different possible determining sequences. By the Pigeonhole principle, this means that among the integers

$$a_k + 1, a_k + 2, \ldots, a_k + 3^{a_k - a_1 + 1} + 1,$$

two of them must have the same determining sequence. □

Let n_1 and n_2 be two integers with the same determining sequence and $n_2 > n_1$. The first claim implies that $f(n_2 + t) = f(n_1 + t)$ for all $t \geq 0$. Setting $p = t + n_1$ gives

$$f(p + (n_2 - n_1)) = f(p)$$

for all $p \geq n_1$. Hence the sequence $\{f(n)\}$ is eventually periodic, from which the desired conclusion follows. □

b. We claim that the set T of prime numbers results in an aperiodic set of good numbers. Clearly every prime number is good, since Alice wins when n is prime by taking all the stones.

Claim: All good numbers are prime.

Proof. Let k be the least nonprime good number. Hence there exists a prime $p < k$ such that $k - p$ is bad. Since 1 and 2 are not bad, we have $k - p \geq 3$. Since $k - p$ is bad, if we let p_m be the greatest prime less than $k - p$, then all of the elements of

$$S = \{k - p - 2, k - p - 3, \ldots, k - p - p_m\}$$

are good. This set must also be nonempty, for otherwise $k - p$ would not be bad. Since each element of S is less than k, the minimal nonprime good number, all elements of S are prime.

If $k - p$ is even, then so is $k - p - 2$. Since S is nonempty we have $k - p - 2 \in S$, and thus $k - p - 2 = 2$, as 2 is the only even prime. Hence $k - p = 4$. However, 4 is neutral, contradicting the badness of $k - p$.

If $k - p$ is odd, then $k - p - 3$ is even. If $k - p > 3$, then $k - p - 3 \in S$, and is thus prime. Therefore,

$$k - p - 3 = 2 \implies k - p = 5.$$

But 5 is good, so we have a contradiction. Finally, we consider when $k - p \leq 3$. However, this implies $k - p \in \{1, 2, 3\}$, none of which are bad numbers (quite fine ones indeed). This is a contradiction, so k cannot exist. Hence all good numbers are prime. □

Claim: The prime numbers are not eventually periodic.

Proof. Suppose the primes are eventually periodic. Since there are infinitely many primes, we can select a prime p such that $p + nk$ is prime for all nonnegative integers n and some fixed positive integer k. Setting $n = p$ gives that $p + pk$ is prime. However, since

$$p + pk = p(1 + k)$$

and both $p, 1 + k > 1$, this is a contradiction. □

We have proved that a number is good if and only if it is prime, and also that the prime numbers are not eventually periodic; we are done. □

Problem 7

Solution: We claim the answer is 10, which can be achieved by setting $(a, b) = (\frac{3}{2}, -\frac{1}{2})$. We prove that there are no smaller solutions.

Suppose that there exists a positive integer $k < 10$, that is not a perfect square, such that $k = \frac{a^2+b^2}{ab+1}$ for some rationals a, b with $ab \neq -1$. Let $a = \frac{p}{q}$ and $b = \frac{r}{s}$, where p, q, r, s are integers, $\gcd(p, q) = \gcd(r, s) = 1$, and $q, s > 0$.

Claim: Both p and r are nonzero.

Proof. Without loss of generality, suppose $p = 0$. Thus $a = 0$, so

$$\begin{aligned} k &= \frac{a^2 + b^2}{ab + 1} \\ &= \frac{b^2}{1} \\ &= b^2 \end{aligned}$$

Thus $\sqrt{k}$ is rational, so k must be a perfect square. But this is a contradiction, so p, and similarly r, must be nonzero. □

Claim: We must have $q = s$.

Proof. We first expand the expression for k.

$$\begin{aligned} k &= \frac{a^2 + b^2}{ab + 1} \\ &= \frac{(p/q)^2 + (r/s)^2}{pr/qs + 1} \\ &= \frac{p^2s^2 + q^2r^2}{qs(pr + qs)} \end{aligned}$$

Now let $d = \gcd(q, s)$, so $q = q_1 d$ and $s = s_1 d$ for some relatively prime positive integers q_1, s_1 with $\gcd(q_1, s_1) = 1$. Suppose for the sake of contradiction that $q > d$, so $q_1 > 1$. We have

$$\begin{aligned} k &= \frac{p^2(s_1 d)^2 + (q_1 d)^2 r^2}{(q_1 d)(s_1 d)(pr + qs)} \\ &= \frac{p^2 s_1^2 + q_1^2 r^2}{q_1 s_1 (pr + qs)}. \end{aligned}$$

Multiplying both sides by the denominator and reducing modulo q_1 gives

$$p^2 s_1^2 \equiv 0 \pmod{q_1}.$$

Since $\gcd(p, q) = 1$, it follows that $\gcd(p^2, q_1) = 1$. Because of this, and the fact that $p \neq 0$, we may divide out p^2 from the congruence. This leaves $s_1^2 \equiv 0$

(mod q_1). However, since $\gcd(q_1, s_1) = 1$, $s_1 \neq 0$, and $q_1 > 1$, this is a contradiction. Therefore, we must have $d \geq q$, and since q is positive by construction, $d = q$. A similar argument with s shows that $d = s$; hence $q = s$. $\square$

As a result, our expression for k simplifies. We have

$$\begin{aligned} k &= \frac{p^2 s^2 + q^2 r^2}{qs(pr + qs)} \\ &= \frac{p^2 q^2 + q^2 r^2}{q^2(pr + q^2)} \\ &= \frac{p^2 + r^2}{pr + q^2}. \end{aligned}$$

Since $0 < k < 10$ and k is not a perfect square, there are only six integers that k could equal: 2, 3, 5, 6, 7, and 8. We consider three cases: $k = 7$, $k \in \{3, 6\}$, and $k \in \{2, 5, 8\}$, and we show that each contradicts $\gcd(p, q) = 1$.

Case 1 ($k = 7$). Here we have

$$\begin{aligned} \frac{p^2 + r^2}{pr + q^2} = 7 &\implies p^2 + r^2 = 7pr + 7q^2 \\ &\implies p^2 + r^2 \equiv 0 \pmod 7. \end{aligned} \tag{2}$$

The quadratic residues modulo 7 are 0, 1, 2, and 4. The only pair of these which sums to 0 (mod 7) is $(0, 0)$. Therefore, we must have

$$p^2 \equiv r^2 \equiv 0 \pmod 7 \implies p \equiv r \equiv 0 \pmod 7.$$

Hence $p = 7p_1$ and $r = 7r_1$ for some integers p_1, r_1. Plugging this into (2) gives

$$49p_1^2 + 49r_1^2 = 343p_1 r_1 + 7q^2 \implies 7p_1^2 + 7r_1^2 = 49p_1 r_1 + q^2.$$

Reducing modulo 7 gives $q^2 \equiv 0 \pmod 7$, so $q = 7q_1$ for some integer q_1. Thus 7 divides both p and q. Since $p, q \neq 0$ and $\gcd(p, q) = 1$, this is a contradiction.

Case 2 ($k \in \{3, 6\}$). This is basically the same as the previous case. We have

$$\begin{aligned} \frac{p^2 + r^2}{pr + q^2} = k &\implies p^2 + r^2 = kpr + kq^2 \\ &\implies p^2 + q^2 \equiv 0 \pmod 3, \end{aligned}$$

since here k is a multiple of 3. The quadratic residues modulo 3 are 0 and 1; the only pair of these that sum to 0 (mod 3) is $(0, 0)$. Thus

$$p^2 \equiv r^2 \equiv 0 \pmod 3 \implies p \equiv r \equiv 0 \pmod 3.$$

Hence $p = 3p_1$ and $r = 3r_1$ for some integers p_1, r_1. Substituting this into the original equation yields

$$9p_1^2 + 9r_1^2 = 9kp_1r_1 + kq^2 \implies 3p_1^2 + 3r_1^2 = 3p_1r_1 + \frac{k}{3}q^2.$$

We take the above equation modulo 3 to obtain

$$\frac{k}{3}q^2 \equiv 0 \pmod 3.$$

Since $\frac{k}{3} \in \{1, 2\}$, we can divide it out of the congruence to obtain $q^2 \equiv 0 \pmod 3$. Thus q must be a multiple of 3, just like p, so we have our contradiction.

Case 3 ($k \in \{2, 5, 8\}$). This case is more interesting. We have

$$\begin{aligned} p^2 + r^2 &= kpr + kq^2 \\ \implies (p - r)^2 &= (k - 2)pr + kq^2. \end{aligned} \tag{3}$$

If $k = 2$, then this equation becomes $(p - r)^2 = 2q^2$, which is impossible by the irrationality of $\sqrt{2}$. So we have $k \in \{5, 8\}$. In this case, $k \equiv 2 \pmod 3$, thus we take (3) modulo 3 to obtain

$$(p - r)^2 \equiv 2q^2 \pmod 3$$

Since 0 and 1 are the only quadratic residues modulo 3, we see that we must have $(p - r)^2 \equiv q^2 \equiv 0 \pmod 3$. Hence $p - r$ and q are multiples of 3, so we have $p - r = 3n$ and $q = 3q_1$ for some integers n, q_1. Substituting these into (3) yields

$$\begin{aligned} 9n^2 &= (k - 2)(3n + r)r + 9kq_1^2 \\ \implies 3n^2 &= (k - 2)nr + \frac{k - 2}{3}r^2 + 3kq_1^2 \end{aligned}$$

Taking this equation modulo 3 yields

$$\frac{k - 2}{3}r^2 \equiv 0 \pmod 3.$$

Since $\frac{k-2}{3} \in \{1, 2\}$, we can divide it out of the congruence to obtain $r^2 \equiv 0 \pmod 3$, so $r \equiv 0 \pmod 3$. Since $p \equiv r \pmod 3$, p is a multiple of three, just like q, giving us our final contradiction. □